

Fractional Cusps at the Atomic Phase

Sharp noninteger regularity of Thue–Morse pressure
and sharp relaxation for digital products in every base

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Abstract

For the doubling map and the singular potential $\psi_c(x) = \log \cos^2 \pi(x - c)$, we determine the phase regularity of $P(q\psi_c)$ at the atomic phase for every $q > 1/2$. For every noninteger q in this range the pressure is locally $C^{\lceil 2q \rceil - 1}$, but its $\lceil 2q \rceil$ th derivative at the atomic phase does not exist. Its leading nonanalytic term is explicit:

$$P(q\psi_c) = 2q \log \cos \pi c + 2(2^{2q} - 1)\zeta(2q)|c|^{2q} + o(|c|^{2q}).$$

Thus analyticity at integer orders is exceptional, not evidence for analyticity at all positive orders.

The proof constructs a base-independent, periodized-sinc eigenfunction and establishes a sharp operator-norm relaxation law on every Hölder space. This supplies the spectral gap directly, without assuming a general Perron theorem for a potential with zeros. A positive-eigenfunction perturbation argument, a pressure–integral comparison, and one exact identity at the fixed point then give the singular expansion. The same argument works in every integer base $b \geq 2$, with cusp coefficient $2(b^{2q} - 1)\zeta(2q)$. At positive integer $q = m$, the would-be cusp cancels exactly the $2m$ th coefficient of the analytic fixed-branch contribution, recovering and extending the repository’s missing-coefficient phenomenon.

The article includes full proofs, a sharpness argument for the convergence rate, finite-size moment asymptotics, numerical diagnostics, explicit provenance and limitations, and a research agenda. The results are claims proved in this research draft; they are not independently refereed or proof-assistant verified.

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1 The question, the answer, and the scope

1.1 A published regularity question

Gohlke, Kesseböhmer, and Schindler prove order-two phase analyticity for this pressure family and leave stronger regularity at general positive orders open [1, Theorem 2.7 and subsequent discussion]. Gohlke, Lamprinakos, and Schmeling subsequently prove continuity for all nonnegative orders and explicitly repeat the stronger-regularity question [2, Theorems 2.2–2.3 and the following paragraph]. The latter paper uses $\log \sin^2 \pi(x - a)$; its atomic phase $a = 1/2$ corresponds to our phase $c = 0$, by $a = c + 1/2 \pmod{1}$.

This article settles the local question at that exceptional phase for all orders strictly above $1/2$. There is both a positive and a negative answer: there are as many continuous derivatives as the order of vanishing of the weight permits, but one further derivative fails at every noninteger order. In particular, global real-analyticity at every positive order is false. We do not claim to classify regularity at all other phases or at orders at or below $1/2$.

1.2 The connection to ProveIt

The inspected repository contains a substantial Fabius/Rvachev and Thue–Morse analysis program [3]. Its recent article *Integer Pressure and a Missing Taylor Coefficient* proves all-positive-integer phase analyticity, a vanishing $2m$ th Taylor coefficient, and a sixth-moment phase optimum [4]. That article explicitly excludes noninteger regularity. The present work addresses this excluded direction rather than repeating its finite-matrix construction or its optimization theorem.

The main observation is that the periodized sinc function appearing at integer orders still makes sense at every real exponent $s > 1$. Its Fourier support need not be finite, but its transfer iterates can be estimated explicitly. The loss of finite Fourier support is therefore not the fundamental obstacle. The surviving fractional power in the fixed-point eigenvalue equation is the actual source of nonanalyticity.

1.3 Status and boundaries of the claim

All central arguments below are supplied in ordinary mathematics. They use no unverified theorem from the repository as an axiom. The repository is credited for the integer-order predecessor, while the new proof rederives the atomic eigenfunction and the coefficient cancellation in a larger real-exponent framework.

The literature check included both primary papers above, the repository’s integer-pressure package, and targeted searches for fractional phase singularities. No matching sharp classification was located in those sources. This is a bounded search, not a guarantee of worldwide priority. Nothing in this package is a Lean or Rocq proof, and numerical agreement is not substituted for a proof of an infinite statement.

Status	Content
Prior work	Positive-order continuity; order-two analyticity; the repository’s integer-order analyticity and missing coefficient.
Proved in this draft	The noninteger atomic cusp and exact integer differentiability class; a sharp Hölder operator-norm relaxation law; the all-base extension; a local uniform finite-size moment expansion.
Not settled here	The critical and subcritical phase asymptotics, regularity at arbitrary nonzero phases, endpoint Hölder regularity on whole neighborhoods, and independent priority or formal verification.

2 Definitions and main results

2.1 Digital masks and pressure

Let $b \geq 2$ be an integer and let $T_b(x) = bx \pmod{1}$ on $\mathbb{T} = \mathbb{R}/\mathbb{Z}$. Define the normalized digital mask

$$d_b(x) = \frac{1}{b} \sum_{r=0}^{b-1} e^{2\pi i r x}, \quad m_b(x) = |d_b(x)| = \left| \frac{\sin \pi b x}{b \sin \pi x} \right|, \quad (2.1)$$

with removable values at integers. For $s > 1$ and a real phase c , put

$$w_{b,s,c}(x) = m_b(x - c)^s, \quad (\mathcal{L}_{b,s,c} f)(x) = \sum_{k=0}^{b-1} w_{b,s,c} \left(\frac{x+k}{b} \right) f \left(\frac{x+k}{b} \right). \quad (2.2)$$

There is no factor $1/b$ in this transfer operator.

Let $\mathcal{D}_{b,n}$ be the b^n closed b -adic intervals of length b^{-n} . Shared endpoints cause no problem for the integrals used below. Set

$$F_{b,s,n,c}(x) = \prod_{j=0}^{n-1} w_{b,s,c}(T_b^j x), \quad (2.3)$$

$$Z_{b,s,n}(c) = \sum_{I \in \mathcal{D}_{b,n}} \sup_{x \in I} F_{b,s,n,c}(x), \quad (2.4)$$

$$P_{b,s}(c) = \lim_{n \rightarrow \infty} \frac{1}{n} \log Z_{b,s,n}(c). \quad (2.5)$$

The existence and spectral interpretation of the last limit near $c = 0$ will be proved, not assumed. All logarithms are natural.

For the binary potential in the introduction,

$$p_q(c) := P(q\psi_c) = P_{2,2q}(c), \quad \psi_c(x) = \log \cos^2 \pi(x - c). \quad (2.6)$$

Thus s is the absolute-moment exponent, while $q = s/2$ is the usual pressure order. This distinction accounts for the exponent $2q$ in the binary theorem.

2.2 The fractional normal form

For $|c| < 1/b$, define the positive analytic function

$$D_b(c) = \frac{\sin \pi b c}{b \sin \pi c}, \quad D_b(0) = 1, \quad C_{b,s} = 2(b^s - 1)\zeta(s). \quad (2.7)$$

Theorem 2.1 (Atomic-phase normal form). *Fix $b \geq 2$ and $s > 1$. There is $\varepsilon > 0$ such that on $|c| < \varepsilon$ the pressure exists, is even, and has the exact form*

$$P_{b,s}(c) = s \log D_b(c) + \log(1 + |c|^s E_{b,s}(c)), \quad (2.8)$$

where $E_{b,s}$ is positive and even, satisfies

$$E_{b,s}(0) = C_{b,s}, \quad (2.9)$$

and is locally C^r for every nonnegative integer $r < s$. Consequently,

$$P_{b,s}(c) = s \log D_b(c) + C_{b,s}|c|^s + o(|c|^{s+1}). \quad (2.10)$$

For even integer s , both the pressure and $E_{b,s}$ are real-analytic near zero.

The one-extra-order remainder in equation (2.10) is useful but is not needed for the nonanalyticity conclusion. It follows from evenness and one continuous derivative of $E_{b,s}$, not from an assumed power series for that function.

Theorem 2.2 (Sharp integer differentiability class). *Let $s > 1$ and assume that s is not an even integer. Put $k = \lceil s \rceil - 1$. Then $P_{b,s}$ is C^k in a neighborhood of zero, but its derivative of order $k + 1$ at zero does not exist. More precisely, for every integer $0 \leq r < s$, as $c \rightarrow 0$ through nonzero real values,*

$$\frac{d^r}{dc^r} (P_{b,s}(c) - s \log D_b(c)) = C_{b,s}(s)_r \operatorname{sgn}(c)^r |c|^{s-r} + o(|c|^{s-r}), \quad (2.11)$$

$$(s)_r = s(s-1) \cdots (s-r+1), \quad (s)_0 = 1. \quad (2.12)$$

Corollary 2.3 (Binary pressure). *For every noninteger $q > 1/2$, p_q is locally $C^{\lceil 2q \rceil - 1}$ but has no derivative of order $\lceil 2q \rceil$ at zero. Its expansion is*

$$p_q(c) = 2q \log \cos \pi c + 2(2^{2q} - 1)\zeta(2q)|c|^{2q} + o(|c|^{2q+1}). \quad (2.13)$$

In particular, noninteger positive orders above $1/2$ do not admit a real-analytic phase dependence on the whole circle.

2.3 The sharp relaxation theorem

Use the circle distance $\delta(x, y)$ and the norm

$$\|f\|_\alpha = \|f\|_\infty + [f]_\alpha, \quad [f]_\alpha = \sup_{x \neq y} \frac{|f(x) - f(y)|}{\delta(x, y)^\alpha}$$

on the complex Banach space $C^\alpha(\mathbb{T})$, where $0 < \alpha \leq 1$. The case $\alpha = 1$ means Lipschitz functions, not continuously differentiable functions. Define

$$H_s(x) = \sum_{\ell \in \mathbb{Z}} |\operatorname{sinc}(x + \ell)|^s, \quad \operatorname{sinc} u = \frac{\sin \pi u}{\pi u}, \quad \operatorname{sinc} 0 = 1, \quad \Pi_s f = f(0)H_s. \quad (2.14)$$

Theorem 2.4 (Sharp atomic relaxation). *For every $b \geq 2$, $s > 1$, and $0 < \alpha \leq 1$, H_s is strictly positive and belongs to $C^1(\mathbb{T})$, and*

$$\mathcal{L}_{b,s,0} H_s = H_s, \quad \mathcal{L}_{b,s,0} \Pi_s = \Pi_s \mathcal{L}_{b,s,0} = \Pi_s. \quad (2.15)$$

There are positive constants a, A , depending only on b, s, α , such that for all $n \geq 1$,

$$aR_{b,s,\alpha}(n) \leq \|\mathcal{L}_{b,s,0}^n - \Pi_s\|_{C^\alpha \rightarrow C^\alpha} \leq AR_{b,s,\alpha}(n), \quad (2.16)$$

where

$$R_{b,s,\alpha}(n) = \begin{cases} b^{-(s-1)n}, & 1 < s < 1 + \alpha, \\ (n+1)b^{-\alpha n}, & s = 1 + \alpha, \\ b^{-\alpha n}, & s > 1 + \alpha. \end{cases} \quad (2.17)$$

In particular, 1 is an algebraically simple isolated eigenvalue and all remaining spectrum lies in $|z| \leq \max\{b^{-\alpha}, b^{1-s}\} < 1$.

The logarithmic factor at $s = 1 + \alpha$ is necessary in this operator norm. It is detected by the single test function $\delta(x, 0)^\alpha$; it is not an artifact of an abstract spectral estimate.

3 The atomic eigenfunction and the telescoping identity

3.1 Construction and regularity of the periodization

For $x \notin \mathbb{Z}$, the series in equation (2.14) becomes

$$H_s(x) = \frac{|\sin \pi x|^s}{\pi^s} \sum_{\ell \in \mathbb{Z}} |x + \ell|^{-s}. \quad (3.1)$$

On compact subsets away from the integers, the series and its first derivative converge uniformly. Near zero, separate the $\ell = 0$ term:

$$H_s(x) = |\operatorname{sinc} x|^s + \frac{|\sin \pi x|^s}{\pi^s} \sum_{\ell \neq 0} |x + \ell|^{-s}. \quad (3.2)$$

The second sum is continuously differentiable there, with uniformly convergent derivative series. Since $s > 1$, multiplication by $|\sin \pi x|^s$ preserves C^1 regularity at zero. Thus $H_s \in C^1$, $H_s(0) = 1$, and periodicity follows by reindexing.

Every x has an integer translate u with $|u| \leq 1/2$. The corresponding summand is at least $(2/\pi)^s$. Hence

$$H_s(x) \geq (2/\pi)^s > 0 \quad (x \in \mathbb{T}). \quad (3.3)$$

This uniform lower bound is important: it allows a small perturbation of the eigenfunction to remain strictly positive.

3.2 The eigenfunction identity

For $y = (x + k)/b$ with $0 \leq k < b$, use $|\sin \pi by| = |\sin \pi x|$ in equation (3.1). Away from removable points,

$$\begin{aligned} m_b(y)^s H_s(y) &= \frac{|\sin \pi x|^s}{b^s \pi^s} \sum_{\ell \in \mathbb{Z}} |y + \ell|^{-s} \\ &= \frac{|\sin \pi x|^s}{\pi^s} \sum_{\ell \in \mathbb{Z}} |x + k + b\ell|^{-s}. \end{aligned}$$

Summing over k partitions $\mathbb{Z}$ into its residue classes modulo b , giving $\mathcal{L}_{b,s,0} H_s = H_s$. Continuity handles the removable points. At $x = 0$, only the inverse branch $y = 0$ has nonzero weight, so

$$(\mathcal{L}_{b,s,0} f)(0) = f(0). \quad (3.4)$$

Together these identities prove the projector relations in theorem 2.4.

3.3 An exact formula for every iterate

Let $N = b^n$. The product of the masks telescopes:

$$\prod_{j=0}^{n-1} m_b(b^j y)^s = \left| \frac{\sin \pi N y}{N \sin \pi y} \right|^s. \quad (3.5)$$

Therefore, for any complete set K_N of N consecutive integer residues modulo N ,

$$(\mathcal{L}_{b,s,0}^n f)(x) = \sum_{k \in K_N} a_{N,k}(x) f\left(\frac{x+k}{N}\right), \quad a_{N,k}(x) = \left| \frac{\sin \pi x}{N \sin(\pi(x+k)/N)} \right|^s. \quad (3.6)$$

Take representatives $x \in [-1/2, 1/2]$ and a centered K_N containing zero, for example $K_N = \{-N/2, \dots, N/2 - 1\}$ when N is even. At the removable zero of $a_{N,0}$, set its value to 1.

This formula is the analytic substitute for a finite moment matrix. The proof below keeps track of both the total mass and its Hölder variation, which is essential for a genuine spectral perturbation argument.

4 Proof of the sharp relaxation law

4.1 Uniform bounds on the branch coefficients

Lemma 4.1. *For fixed $s > 1$, there is a constant C_s independent of $N \geq 2$ such that on $[-1/2, 1/2]$,*

$$\|a_{N,0}\|_{C^1} \leq C_s, \quad \|a_{N,k}\|_{C^1} \leq C_s |k|^{-s} \quad (0 \neq k \in K_N). \quad (4.1)$$

Here C^1 is used only for the scalar coefficient functions on the interval.

Proof. For $k = 0$,

$$a_{N,0}(x) = \left| \frac{\text{sinc } x}{\text{sinc}(x/N)} \right|^s.$$

Both factors are uniformly bounded away from zero on the relevant intervals, and their first derivatives are uniformly bounded.

For $k \neq 0$, one has $|x + k| \geq |k|/2$. The quantity $(x + k)/N$ stays in a fixed interval strictly inside $(-1, 1)$, and thus

$$N |\sin(\pi(x + k)/N)| \geq c_0 |x + k| \geq (c_0/2) |k|$$

with an absolute positive c_0 . Differentiating the quotient, the derivative of the numerator power is uniformly bounded because $s > 1$. The logarithmic derivative of the denominator is bounded by $C/|x + k|$:

$$\left| \frac{\pi}{N} \cot \frac{\pi(x + k)}{N} \right| \leq \frac{C}{|x + k|}.$$

These estimates give equation (4.1). □

4.2 Upper bound on the invariant ideal

Let

$$\mathcal{I}_\alpha = \{f \in C^\alpha(\mathbb{T}) : f(0) = 0\}.$$

By equation (3.4), this is an invariant closed subspace. For $f \in \mathcal{I}_\alpha$,

$$|f(y)| \leq [f]_\alpha \delta(y, 0)^\alpha. \quad (4.2)$$

Put $f_{N,k}(x) = f((x + k)/N)$. On the representative interval,

$$[f_{N,k}]_\alpha \leq N^{-\alpha} [f]_\alpha, \quad (4.3)$$

$$\|f_{N,0}\|_\infty \leq C N^{-\alpha} [f]_\alpha, \quad (4.4)$$

$$\|f_{N,k}\|_\infty \leq C N^{-\alpha} |k|^\alpha [f]_\alpha \quad (k \neq 0). \quad (4.5)$$

The estimates remain valid when an inverse image crosses a chosen representative endpoint, because f is periodic and circle distance is at most the distance between its lifts.

Using the product inequality $[uv]_\alpha \leq \|u\|_\infty [v]_\alpha + [u]_\alpha \|v\|_\infty$, lemma 4.1, and equation (4.5), we obtain

$$\|\mathcal{L}_{b,s,0}^n f\|_\alpha \leq C N^{-\alpha} \left(1 + \sum_{k=1}^N k^{\alpha-s} \right) \|f\|_\alpha. \quad (4.6)$$

For completeness, the interval estimate gives the circle estimate up to a fixed factor: for two points separated by the cut, split their difference at the identified endpoints and use $a^\alpha + b^\alpha \leq 2^{1-\alpha}(a+b)^\alpha$.

The elementary power-sum alternatives are

$$1 + \sum_{k=1}^N k^{\alpha-s} \asymp \begin{cases} N^{1+\alpha-s}, & s < 1 + \alpha, \\ 1 + \log N, & s = 1 + \alpha, \\ 1, & s > 1 + \alpha. \end{cases} \quad (4.7)$$

Since $N = b^n$, equation (4.6) gives the upper rate in equation (2.16) on $\mathcal{I}_\alpha$. For arbitrary f , write

$$f = f(0)H_s + g, \quad g \in \mathcal{I}_\alpha, \quad \|g\|_\alpha \leq C_s \|f\|_\alpha.$$

Because H_s is fixed by the operator, the same bound holds for $\mathcal{L}^n - \Pi_s$ on the whole space.

4.3 Matching lower bound

Lemma 4.2. *The fixed function $f_\alpha(x) = \delta(x, 0)^\alpha$ satisfies*

$$(\mathcal{L}_{b,s,0}^n f_\alpha)(1/2) \geq cN^{-\alpha} \left(1 + \sum_{k=1}^{\lfloor N/4 \rfloor} k^{\alpha-s} \right) \quad (4.8)$$

for all sufficiently large $N = b^n$.

Proof. At $x = 1/2$, the numerator of $a_{N,k}$ has absolute value 1. For $0 \leq k \leq N/4$, the inverse image $(k + 1/2)/N$ lies in $[0, 1/2]$, so its distance to zero is exactly $(k + 1/2)/N$. Also,

$$N \sin \frac{\pi(k + 1/2)}{N} \leq \pi(k + 1/2).$$

Consequently each such summand in equation (3.6) is at least

$$\pi^{-s} N^{-\alpha} (k + 1/2)^{\alpha-s}.$$

All summands are nonnegative, and the $k = 0$ term supplies the initial constant in equation (4.8). $\square$

The function f_α belongs to C^α with bounded norm and $f_\alpha(0) = 0$. Thus $\Pi_s f_\alpha = 0$. Applying equation (4.7) to equation (4.8) gives the lower operator-norm bound. Changing constants covers the finitely many small n . This proves the two-sided estimate in theorem 2.4.

4.4 The spectral consequence

Write $R = \mathcal{L}_{b,s,0} - \Pi_s$. The projector relations imply $R^n = \mathcal{L}_{b,s,0}^n - \Pi_s$ for $n \geq 1$, and the estimates give

$$r(R) \leq \max\{b^{-\alpha}, b^{1-s}\} < 1.$$

Since the range of Π_s is one-dimensional, this decomposes the operator into the simple eigenvalue 1 and a strictly smaller spectral part. In particular there is no generalized eigenvector attached to 1. This supplies the spectral gap needed below directly, rather than by invoking a positivity theorem that might fail at an atomic phase.

5 Parameter regularity and spectral perturbation

5.1 A translation lemma on Hölder spaces

Lemma 5.1. *Fix $s > 1$ and a nonnegative integer $r < s$. Choose $0 < \alpha < \min\{1, s - r\}$. Then*

$$c \mapsto w_{b,s,c} \in C^\alpha(\mathbb{T}), \quad c \mapsto \mathcal{L}_{b,s,c} \in \mathcal{B}(C^\alpha(\mathbb{T}))$$

are C^r maps. If s is an even integer, the maps are real-analytic.

Proof. The nontrivial zeros of the mask occur at j/b , $1 \leq j < b$, and are simple zeros before taking the absolute value. In a local coordinate u around each zero, the weight has the form $|u|^s v(u)$ with v positive and real-analytic. Away from those zeros it is real-analytic. Its r th derivative is Hölder continuous of every exponent $\gamma < \min\{1, s - r\}$. Choose such a γ larger than α .

Translations of a C^γ function are continuous in the C^α norm. One elementary proof splits the Hölder quotient into distances at most η and greater than η : the first part is bounded by $C\eta^{\gamma-\alpha}$ and the second by $2\eta^{-\alpha}$ times the uniform translation error. First choose η , then let the translation tend to zero. Applying this to the r th derivative, and using the integral form of the Taylor remainder for the lower derivatives, proves C^r translation dependence in C^α .

Multiplication is bounded on C^α , and the finite sum over inverse branches is bounded there. Hence the same parameter regularity holds for the operator. When $s = 2m$, the weight is a trigonometric polynomial, so its translation depends analytically on c in the operator norm. $\square$

The strict inequality $\alpha < s - r$ is intentional. Translating a function at its exact Hölder endpoint need not be norm-continuous. Passing to a slightly weaker Hölder space avoids that pitfall without losing an integer derivative of the pressure.

5.2 The isolated eigenvalue persists

Proposition 5.2. *For every fixed $b \geq 2$, $s > 1$, and integer $r < s$, a neighborhood of zero has C^r functions $\rho(c) > 0$ and $h_c \in C^\alpha(\mathbb{T})$ such that*

$$\mathcal{L}_{b,s,c} h_c = \rho(c) h_c, \quad h_c(0) = 1, \quad h_c > 0, \quad \rho(0) = 1, \quad h_0 = H_s. \quad (5.1)$$

The eigenvalue is algebraically simple and strictly dominant. There is also a C^r rank-one spectral projector Π_c , and, after possibly shrinking the neighborhood,

$$\|\rho(c)^{-n} \mathcal{L}_{b,s,c}^n - \Pi_c\|_{\alpha \rightarrow \alpha} \leq K\theta^n \quad (n \geq 0) \quad (5.2)$$

with $0 < \theta < 1$ and $K < \infty$ independent of c . For even integer s , the spectral data are real-analytic.

Proof. By theorem 2.4, a small circle Γ around 1 separates that simple eigenvalue from the rest of the spectrum of $\mathcal{L}_{b,s,0}$. Operator-norm continuity and the Neumann series preserve the resolvent on Γ for small c . Define

$$\Pi_c = \frac{1}{2\pi i} \int_{\Gamma} (z - \mathcal{L}_{b,s,c})^{-1} dz.$$

Differentiating the resolvent identity gives the stated C^r dependence. The projectors remain rank one: sufficiently close projections have isomorphic ranges, by restricting one projector to the other's range. Since $\Pi_0 1 = H_s$, set

$$h_c = \frac{\Pi_c 1}{(\Pi_c 1)(0)}.$$

The denominator tends to 1, and equation (3.3) ensures that h_c remains strictly positive. The invariant one-dimensional range supplies an eigenvalue; evaluating at zero gives $\rho(c) = (\mathcal{L}_{b,s,c} h_c)(0)$. Reality, positivity, and C^r regularity follow immediately.

The remaining spectrum stays in a disk of radius less than 1, whereas $\rho(c)$ tends to 1. This can be seen directly from resolvent stability: outside fixed neighborhoods of the old spectrum, cover a bounded region by finitely many resolvent neighborhoods and use the Neumann series; large $|z|$ are handled by the uniform operator-norm bound. Thus the eigenvalue remains strictly dominant.

For the uniform estimate, normalize the complementary operator by $\rho(c)$. A fixed circle $|z| = \theta < 1$ encloses all of its spectra for small c , with uniformly bounded resolvents on that circle. The Cauchy integral formula for its powers gives equation (5.2). Enlarge K to include $n = 0$. Analytic input gives analytic resolvents and spectral data when s is even. $\square$

Although different r may require different Hölder spaces, they give the same positive eigenvalue and normalized positive eigenfunction near zero. To see uniqueness without comparing the Banach spaces, compare two strictly positive eigenfunctions by their maximum and minimum ratio. Positivity of $\mathcal{L}$ forces their positive eigenvalues to be equal; their difference then lies in the unique eigenspace in a common weaker Hölder space. Hence the resulting pressure regularities are compatible.

5.3 Reflection symmetry

Let $(Jf)(x) = f(-x)$. Evenness of the mask gives

$$J\mathcal{L}_{b,s,c} = \mathcal{L}_{b,s,-c}J.$$

Uniqueness and the normalization at zero imply

$$\rho(-c) = \rho(c), \quad h_{-c}(x) = h_c(-x). \quad (5.3)$$

The atomic eigenfunction is therefore the common limit of the perturbed positive eigenfunctions from both sides.

6 Why the spectral eigenvalue is the pressure

A positive eigenvalue on a chosen function space does not by itself identify cylinder-supremum pressure. The following comparison closes that gap.

Lemma 6.1 (A polynomial variation bound). *For every $p \geq 1$ there is a constant B_p such that every trigonometric polynomial Q of degree at most $D \geq 1$ satisfies*

$$\int_0^1 \left| \frac{d}{dx} |Q(x)|^p \right| dx \leq B_p D \int_0^1 |Q(x)|^p dx. \quad (6.1)$$

The derivative is interpreted almost everywhere when necessary.

Proof. A convolution proof of the elementary estimate $\|Q'\|_p \leq CD\|Q\|_p$ is given in section A. For $p > 1$, the chain rule and Hölder's inequality give

$$\int |(|Q|^p)'| \leq p \int |Q|^{p-1} |Q'| \leq p \|Q\|_p^{p-1} \|Q'\|_p.$$

For $p = 1$, absolute continuity and $|(|Q|)'| \leq |Q'|$ almost everywhere give the same conclusion. $\square$

Define the digital product

$$Q_{b,n,c}(x) = \prod_{j=0}^{n-1} \left(\sum_{r=0}^{b-1} e^{2\pi i r(b^j x - c)} \right). \quad (6.2)$$

Its degree in $e^{2\pi i x}$ is $b^n - 1$, and

$$F_{b,s,n,c}(x) = b^{-sn} |Q_{b,n,c}(x)|^s. \quad (6.3)$$

On an interval of length $1/N$, the supremum of a nonnegative absolutely continuous function is at least its average and at most its average plus its variation. Summing over the partition, with $N = b^n$, gives

$$N \int F \leq Z_{b,s,n}(c) \leq N \int F + \int |F'| \leq (1 + B_s) N \int F. \quad (6.4)$$

The constants do not depend on n or the phase.

Changing variables in the inverse-branch sums yields

$$\int_0^1 \mathcal{L}_{b,s,c}^n 1 \, dx = N \int_0^1 F_{b,s,n,c}(x) \, dx. \quad (6.5)$$

By proposition 5.2, the left side equals

$$\rho(c)^n (\mathcal{A}_{b,s}(c) + O(\theta^n)), \quad \mathcal{A}_{b,s}(c) = \int_0^1 \Pi_c 1 \, dx.$$

At zero, $\mathcal{A}_{b,s}(0) = \int H_s > 0$. Continuity gives a uniform positive lower bound on a smaller neighborhood. Taking logarithms in equation (6.4) proves both existence of the limit and

$$P_{b,s}(c) = \log \rho(c). \quad (6.6)$$

No bounded-distortion estimate for the logarithm of a vanishing weight has been used.

7 The fixed-point identity and the fractional cusp

7.1 An exact scalar equation

Evaluate the eigenfunction equation at $x = 0$. The principal inverse branch gives $D_b(c)^s$, since $h_c(0) = 1$. Every other inverse branch has numerator $|\sin \pi b c|^s$. Hence

$$\rho(c) = D_b(c)^s + \frac{|\sin \pi b c|^s}{b^s} \sum_{k=1}^{b-1} \frac{h_c(k/b)}{|\sin \pi(k/b - c)|^s}. \quad (7.1)$$

Choose the phase neighborhood small enough that the denominators do not vanish. Write

$$B_{b,s}(c) = \sum_{k=1}^{b-1} \frac{h_c(k/b)}{|\sin \pi(k/b - c)|^s}. \quad (7.2)$$

Dividing equation (7.1) by $D_b(c)^s$ gives the particularly simple identity

$$\rho(c) = D_b(c)^s (1 + |\sin \pi c|^s B_{b,s}(c)). \quad (7.3)$$

Thus the logarithmic singularity of the original potential has become an explicit fractional power in a scalar eigenvalue equation.

The function $B_{b,s}$ is positive and C^r for every integer $r < s$. By equation (5.3), replacing k by $b - k$ shows that it is even. Define

$$E_{b,s}(c) = \left(\frac{\sin \pi c}{c} \right)^s B_{b,s}(c), \quad E_{b,s}(0) = \pi^s B_{b,s}(0), \quad (7.4)$$

where the quotient is positive in the small neighborhood under consideration. Its power is real-analytic there. Equations (6.6) and (7.3) now give the exact normal form.

7.2 Evaluation of the cusp coefficient

Using equation (3.1),

$$\pi^s B_{b,s}(0) = \sum_{k=1}^{b-1} \sum_{\ell \in \mathbb{Z}} |k/b + \ell|^{-s} \quad (7.5)$$

$$= b^s \sum_{\substack{j \in \mathbb{Z} \\ b \nmid j}} |j|^{-s} = 2(b^s - 1)\zeta(s). \quad (7.6)$$

All rearrangements are justified by absolute convergence, since $s > 1$. The same coefficient works in every base, with the base entering only through the omitted multiples in the final zeta sum.

Since $E_{b,s}$ is even and C^1 , its derivative at zero is zero, so $E_{b,s}(c) = C_{b,s} + o(|c|)$. Consequently,

$$\begin{aligned} \log(1 + |c|^s E_{b,s}(c)) &= C_{b,s} |c|^s + o(|c|^{s+1}) + O(|c|^{2s}) \\ &= C_{b,s} |c|^s + o(|c|^{s+1}), \end{aligned}$$

because $s > 1$. This proves theorem 2.1, including its stronger remainder.

7.3 Differentiation and the exact loss of smoothness

Fix an integer $r < s$. On either side of zero, differentiate equation (2.8). The leading product term is

$$C_{b,s} \frac{d^r}{dc^r} |c|^s = C_{b,s}(s)_r \operatorname{sgn}(c)^r |c|^{s-r}.$$

Every term with a positive number j of derivatives falling on E has an additional factor $|c|^j$ relative to this scale and a bounded derivative $E^{(j)}$. The term with no derivative falling on E has $E(c) - C_{b,s} = o(1)$. Finally, the nonlinear part of $\log(1 + u) - u$ contributes $O(|c|^{2s-r})$ under these differentiations. This proves equation (2.11).

Let $k = \lceil s \rceil - 1$ and subtract the analytic function $a(c) = s \log D_b(c)$. The derivatives of $P - a$ up to order k exist continuously and vanish at zero. If s is not an integer, then $0 < s - k < 1$, so

$$\frac{(P - a)^{(k)}(c) - (P - a)^{(k)}(0)}{c} = C_{b,s}(s)_k \operatorname{sgn}(c)^{k-1} |c|^{s-k-1} + o(|c|^{s-k-1})$$

has no finite limit. If s is an odd integer, $k = s - 1$ is even and the two one-sided limits are $\pm C_{b,s} s!$. Thus a derivative of order $k + 1$ fails in both cases. The only remaining integer case is even s , which is analytic and is treated next. This proves theorem 2.2 and corollary 2.3.

8 Integer resonance and explicit consequences

8.1 Why the integer coefficient disappears

Euler's sine product, followed by an absolutely convergent logarithmic expansion, gives for $|c| < 1/b$

$$\log D_b(c) = - \sum_{j \geq 1} \frac{(b^{2j} - 1)\zeta(2j)}{j} c^{2j}. \quad (8.1)$$

For $s = 2m$, equation (2.8) is analytic. At degree $2m$, the first term contributes

$$-\frac{2m}{m} (b^{2m} - 1)\zeta(2m) = -2(b^{2m} - 1)\zeta(2m),$$

and the second contributes exactly the opposite number, by equation (7.6). Hence:

Corollary 8.1 (All-base integer cancellation). *For every integer $m \geq 1$ and base $b \geq 2$,*

$$[c^{2m}]P_{b,2m}(c) = 0, \quad (8.2)$$

and for $1 \leq j < m$,

$$[c^{2j}]P_{b,2m}(c) = -\frac{2m}{j}(b^{2j} - 1)\zeta(2j). \quad (8.3)$$

The binary cancellation is credited to the repository predecessor [4]; the role here is to explain it as the even-integer specialization of a real-exponent normal form and extend it to arbitrary integer bases. At a noninteger exponent, there is no analytic coefficient of degree s available to cancel $|c|^s$.

8.2 The order-one normalization

When $s = 2$, the pressure vanishes identically, not merely to a finite Taylor order. Indeed, discrete Fourier orthogonality gives

$$\sum_{k=0}^{b-1} \left| \frac{1}{b} \sum_{r=0}^{b-1} e^{2\pi i r((x+k)/b-c)} \right|^2 = 1. \quad (8.4)$$

To check it, expand the square and sum over k . Only $r = t$ survives, since $|r - t| < b$. Thus $\mathcal{L}_{b,2,c}1 = 1$ for every phase. The pressure comparison in section 6 gives $P_{b,2}(c) = 0$ everywhere. This is consistent with the normal form and isolates the simplest integer resonance.

8.3 Local sign and curvature

For $1 < s < 2$, the fractional term dominates the analytic quadratic term, so

$$P_{b,s}(c) \sim C_{b,s}|c|^s > 0 \quad (c \rightarrow 0, c \neq 0). \quad (8.5)$$

For $s > 2$, the quadratic term dominates:

$$P_{b,s}(c) = -\frac{s(b^2 - 1)\pi^2}{6}c^2 + o(c^2), \quad P''_{b,s}(0) = -\frac{s(b^2 - 1)\pi^2}{3}. \quad (8.6)$$

Thus the atomic phase is a strict local minimum for $1 < s < 2$, a flat normalization point for $s = 2$, and a strict local maximum for $s > 2$. These assertions are local; no global optimization theorem is inferred.

For the binary family, several representative cases are:

q	Leading fractional coefficient	Exact local integer class
$3/4$	$2(2^{3/2} - 1)\zeta(3/2)$	C^1 , no second derivative at 0
$5/4$	$2(2^{5/2} - 1)\zeta(5/2)$	C^2 , no third derivative at 0
$3/2$	$14\zeta(3)$	C^2 , no third derivative at 0
$7/4$	$2(2^{7/2} - 1)\zeta(7/2)$	C^3 , no fourth derivative at 0
2	The degree-four term cancels	Real-analytic near 0

For example,

$$p_{3/2}(c) = -\frac{3}{2}\pi^2 c^2 + 14\zeta(3)|c|^3 + o(|c|^3).$$

The odd absolute power is a genuine two-sided cusp even though its exponent is an integer.

9 Finite-size moments and the critical boundary

9.1 Uniform asymptotics for finite digital products

Corollary 9.1. *For fixed $b \geq 2$ and $s > 1$, there is a phase neighborhood of zero on which*

$$\int_0^1 |Q_{b,n,c}(x)|^s dx = \mathcal{A}_{b,s}(c) \exp(n((s-1)\log b + P_{b,s}(c))) (1 + O(\theta^n)), \quad (9.1)$$

uniformly in the phase, where $0 < \theta < 1$ and $\mathcal{A}_{b,s}(c) > 0$. The amplitude is C^r for each integer $r < s$, and

$$\mathcal{A}_{b,s}(0) = \int_0^1 H_s(x) dx = \int_{\mathbb{R}} |\operatorname{sinc} u|^s du. \quad (9.2)$$

Proof. Combine equations (5.2), (6.3) and (6.5). The leading coefficient stays bounded away from zero, so an absolute error may be written as a relative error. The final equality follows from Tonelli's theorem applied to the nonnegative periodization; the integral converges because $s > 1$. $\square$

At $c = 0$, the finite product is simply $\sum_{r=0}^{b^n-1} e^{2\pi i r x}$. Away from zero, it is a phase-deformed digit product; for $b = 2$, $c = 1/2$, its coefficients are the classical Thue–Morse signs. The theorem is local near the atomic phase and does not supply a uniform estimate reaching $c = 1/2$.

For binary Riesz-product measures, the established pressure– L^q spectrum identity identifies the two quantities up to division by $\log 2$ at nonzero phases [1, Theorem 2.6]. At orders $q > 1/2$, the pressure tends to zero at the atomic phase, consistent with the trivial spectrum of the point mass. Consequently the same local cusp classification transfers to that phase-dependent spectrum, with its coefficient divided by $\log 2$. This corollary uses the cited measure-theoretic theorem; the pressure proof above does not.

9.2 Why the threshold is not a technical convenience

The coefficient $C_{b,s}$ diverges as $s \downarrow 1$, and the slow relaxation factor b^{1-s} tends to 1. Both facts warn against passing to the critical exponent inside the supercritical expansion. There is also a direct critical obstruction.

Proposition 9.2 (Critical growth at the atomic phase). *Extend the definition of the continuous weight to $s = 1$. Then*

$$\|\mathcal{L}_{b,1,0}^n 1\|_{\infty} \asymp n + 1. \quad (9.3)$$

In particular, the rank-one bounded-power convergence used above cannot hold at the critical exponent.

Proof. The scalar iterate formula equation (3.6) is still valid. Its $k = 0$ coefficient is uniformly bounded, and the remaining coefficients are bounded by $C/|k|$. Hence the supremum is at most $C(1 + \log N)$. At $x = 1/2$, the terms with $1 \leq k \leq N/4$ are at least $1/(\pi(k + 1/2))$, giving a matching lower bound. Use $N = b^n$. $\square$

This proposition does not determine the phase asymptotic of $P_{b,1}(c)$. It identifies the failure of the precise mechanism used for $s > 1$ and motivates a separate critical analysis rather than an unjustified continuation of the zeta coefficient.

10 Computation, reproducibility, and verification limits

10.1 What the supplied program checks

The executable code `verify.py` carries out three independent kinds of finite checks. It evaluates the atomic eigenfunction identity at 65-decimal-digit precision using

$$H_s(x) = \left(\frac{\sin \pi x}{\pi} \right)^s (\zeta(s, x) + \zeta(s, 1 - x)), \quad 0 < x < 1, \quad (10.1)$$

it verifies the residue-class formula for $C_{b,s}$, and it checks the integer cancellation after factoring out the common zeta value using exact symbolic arithmetic. It also iterates a positive piecewise-linear collocation of the transfer operator to test the cusp numerically.

The recorded run contains 112 high-precision eigenfunction checks and 28 high-precision amplitude checks for bases 2, 3, 5, 7 and seven real exponents. Their largest observed absolute residuals are approximately 1.2×10^{-64} and 3.9×10^{-61} , respectively. The symbolic cancellation check covers $m = 1, \dots, 20$ with symbolic base b . These computations test formulas; the all-exponent and all-base results are supplied by the proofs, not inferred from those finite ranges.

10.2 A stable numerical observable

Directly subtracting $s \log D_b(c)$ from a numerical pressure loses precision at large s and small c . Instead the program evaluates the normal-form excess with $\log 1p$:

$$\mathcal{R}_{b,s}(c) = \frac{P_{b,s}(c) - s \log D_b(c)}{|c|^s} = \frac{\log(1 + |\sin \pi c|^s B_{b,s}(c))}{|c|^s}. \quad (10.2)$$

The theorem predicts $\mathcal{R}_{b,s}(c) \rightarrow C_{b,s}$. The following values use the default grid, rounded for display.

b	s	$c = 10^{-2}$	$c = 3 \cdot 10^{-3}$	$c = 10^{-3}$	$C_{b,s}$
2	1.5	8.909155	9.418009	9.522764	9.553076
2	2.5	12.609818	12.505178	12.495476	12.494221
2	3.0	17.065680	16.850219	16.831182	16.828797
2	3.5	23.687926	23.281687	23.246062	23.241609
3	1.5	19.960605	21.496843	21.827067	21.923850
3	2.5	39.702426	39.194272	39.146656	39.140459

The program records 18 such phase tests and two doubled-grid checks. The grid has 65,536 points for base two and is rounded upward to a multiple of the base when necessary. It uses nonnegative linear interpolation, preserving positivity of the discretized operator. The iteration is normalized by its value at zero, matching the proof's eigenfunction normalization.

10.3 What these numbers do not certify

The floating-point residual is the residual of the discretized iteration, not an enclosure of the true operator spectrum. No directed-rounding interval certificate or rigorous discretization-error bound is claimed. The doubled-grid tests are stability diagnostics only. The high-precision zeta checks are also numerical rather than exact proofs; only the separate symbolic cancellation identities use exact arithmetic.

The package deliberately separates these layers in `results/verification.json`. It also includes a source ledger, rebuild instructions, and a claims ledger. No repository file has been modified by preparing this package.

11 Further research questions

Research question 1 (Critical phase asymptotics). Determine the leading behavior of $P_{b,1}(c)$ as $c \rightarrow 0$. The linear critical growth in proposition 9.2, the pole of $\zeta(s)$, and the closing spectral gap should be treated together. An induced or renewal operator may replace the lost rank-one limit. A guessed logarithmic law is not asserted here.

Research question 2 (The subcritical range). For $0 < s < 1$, determine the phase regularity and the correct small-phase expansion. The periodized H_s diverges, and the atomic fixed branch is no longer described by the positive eigenfunction used here. The appropriate normalization and dominant spectral component must be identified before perturbing.

Research question 3 (Regularity at nonzero phases). Classify phase regularity away from the atomic point. Does the answer depend on the orbit of the zero of the weight, for example on its periodicity or recurrence? The atomic result rules out a global analytic theorem, but it does not rule out stronger regularity on substantial subsets of the punctured phase circle.

Research question 4 (A complete fractional expansion). Determine the higher-order structure of $E_{b,s}(c)$. Does the pressure admit a convergent or asymptotic expansion involving even integer powers, $|c|^s$, their products, and possible resonant logarithms? The exact normal form isolates the unknown function and should be a better starting point than differentiating the singular potential directly.

Research question 5 (Endpoint Hölder regularity). For $k = \lceil s \rceil - 1$ and non-even s , determine whether $P_{b,s} \in C^{k,s-k}$ on a full neighborhood when s is nonintegral, and whether it belongs to $C^{s-1,1}$ when s is odd. The derivative asymptotic proves the pointwise order at zero, but a two-point endpoint estimate needs additional control of $E_{b,s}$.

Research question 6 (A uniform critical crossover). Develop a two-parameter asymptotic as $s \downarrow 1$ and $c \rightarrow 0$ simultaneously. The fixed- s theorem is not uniform in that limit. A useful result would identify the correct combined scaling variable and match the supercritical cusp to the critical law.

Research question 7 (The full atomic spectrum). The relaxation theorem determines the sharp norm-decay scale. Determine the complete spectrum on C^α and on smoother or weighted spaces, including the mechanism behind the logarithmic factor at $s = 1 + \alpha$. Distinguish genuine eigenvalues, essential spectrum, and endpoint norm effects rather than inferring one from the other.

Research question 8 (Other masks and multiple zeros). Generalize the method to normalized trigonometric masks whose zeros have different multiplicities, and to several forbidden inverse branches. Which hypotheses yield a computable positive atomic eigenfunction, and when is the first nonanalytic exponent determined by a zero multiplicity? The arithmetic zeta sum in this paper may be replaced by a more general lattice or orbit sum.

Research question 9 (Certified numerics and formalization). Turn the explicit branch bounds into quantitative constants and combine them with interval arithmetic to certify a usable phase neighborhood and finite-size error. A proof-assistant development could then verify the periodization, the sharp power-sum estimate, the spectral separation, and the exact scalar identity in separate modules. This would convert the ordinary-mathematics theorem into the repository's trusted setting.

A A self-contained coarse Bernstein estimate

Here is the convolution argument used in lemma 6.1. Let

$$F_M(x) = \frac{1}{M} \left(\frac{\sin \pi M x}{\sin \pi x} \right)^2$$

be the Fejér kernel. Its Fourier coefficients are $(1 - |k|/M)_+$, as follows by expanding the squared geometric sum. On $|x| \leq 1/2$, direct differentiation gives

$$|F'_M(x)| \leq \begin{cases} CM^2, & |x| \leq M^{-1}, \\ C(|x|^{-2} + M^{-1}|x|^{-3}), & M^{-1} < |x| \leq 1/2. \end{cases} \quad (\text{A.1})$$

For the first bound one may differentiate the finite Fourier expansion: $\sum_{|k| < M} |k|(1 - |k|/M) = O(M^2)$. The second follows from the quotient formula and $|\sin \pi x| \geq 2|x|$ on this interval. Integrating shows $\|F'_M\|_1 \leq CM$.

The de la Vallée Poussin kernel $V_D = 2F_{2D} - F_D$ has Fourier coefficient 1 for every $|k| \leq D$. Hence $Q = Q * V_D$ for every trigonometric polynomial of degree at most D , and

$$\|Q'\|_p = \|Q * V'_D\|_p \leq \|Q\|_p \|V'_D\|_1 \leq CD \|Q\|_p \quad (p \geq 1)$$

by the integral triangle inequality. Only a constant proportional to D is needed for pressure, so no sharp Bernstein constant is required.

B Proof dependencies and a formalization route

The central dependency chain is short:

$$\begin{aligned} & \text{telescoping masks} \implies \text{sharp relaxation} \\ & \implies \text{simple eigenvalue perturbation} \implies \text{fixed-point normal form.} \end{aligned}$$

The polynomial variation bound is a separate bridge identifying the spectral eigenvalue with the cylinder definition of pressure. The zeta sum evaluates the coefficient after the normal form is established. The integer cancellation and the noninteger differentiability theorem then follow from one-variable calculus.

A practical Lean integration could use the following boundaries. First formalize the finite mask and iterate identities, keeping the removable origin explicit. Next prove summability and the C^1 gluing of H_s . Then formalize the invariant ideal estimate with an explicit finite power sum; the lower bound uses only nonnegative summands of a single test function. The spectral step should state and prove a general isolated-rank-one perturbation lemma on a complex Banach space, rather than admitting a specialized Perron theorem. Finally, package the scalar identity at zero and the pressure comparison as independent interfaces.

None of these modules is supplied as already checked code. The present package is designed to expose the theorem boundaries and the delicate points: Hölder translation at a strict exponent, uniform positivity of the eigenfunction, pressure identification in the presence of zeros, and two-sided nonexistence of the next derivative.

C Source and convention ledger

The repository was inspected at commit `afb2d1227d8bc5df3beffc1463e958db3544960b` on 29 September 2026. The main predecessor is located under

```
Analysis/FabiusFunction/docs/semi-formalized-research-frontiers/drafts/
thue-morse/Thue_Morse_Integer_Pressure/
```

The primary-source check used `arXiv:2509.22109v1` and `arXiv:2603.19001v1`. The latter's page 4, including the stronger-regularity question, was inspected in the rendered PDF. A separate machine-readable ledger records URLs, scope, and the snapshot.

The essential convention conversions are

$$s = 2q, \quad p_q(c) = P_{2,2q}(c), \quad a_{\text{sine}} = c_{\text{cosine}} + \frac{1}{2} \pmod{1}.$$

The transfer operator is an unnormalized sum, and the pressure uses natural logarithms. The finite moment growth includes the additional term $(s - 1) \log b$ in equation (9.1). These conventions prevent both the half-period ambiguity and the missing factor of $\log b$.

References

- [1] P. Gohlke, M. Kesseböhmer, and T. I. Schindler, *Generalized Thue–Morse measures: spectral and fractal analysis*, 2025, [arXiv:2509.22109v1](#).
- [2] P. Gohlke, G. Lamprinakos, and J. Schmeling, *On a family of singular potentials: Parameter dependence of thermodynamic characteristics*, 2026, [arXiv:2603.19001v1](#).
- [3] V. Reshetnikov, *ProveIt*, repository snapshot [afb2d1227d8bc5df3beffc1463e958db3544960b](#), inspected 29 September 2026. [Repository snapshot](#).
- [4] ProveIt research corpus, *Integer Pressure and a Missing Taylor Coefficient: Analytic phase dependence and the sixth-moment optimum for generalized Thue–Morse products*, research draft, 28 September 2026. [Pinned LaTeX source](#).